

WORKOUT INTENSITY & CALORIE BURN ANALYSIS

How Exercise Type and Body Composition Influence Fitness Outcomes

Background: This project uses lifestyle data that tracks individual workouts. Each entry tracks demographic information (age, gender), physical characteristics (weight, height, BMI), workout details (type, duration, calories), and heart rate metrics (max, average, and resting BPM). Our goal is to understand how workout intensity is affected by physical characteristics and exercise type

20000
PARTICIPANTS

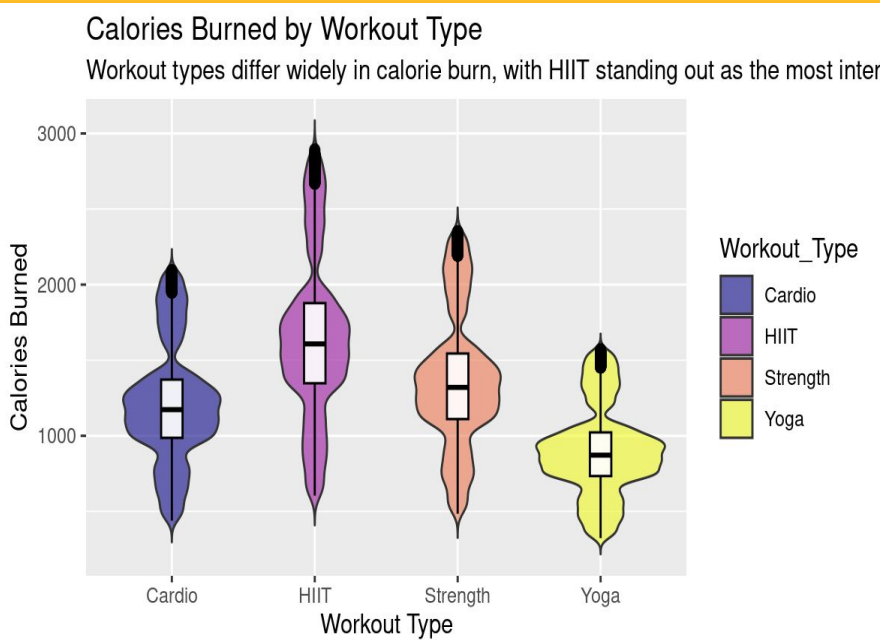
4
WORKOUT
TYPES

11
VARIABLES
TRACKED

FINDING 1: HIIT BURNS MOST CALORIES

FINDING 2: SIMILAR TRENDS; GENDER PEAKS ARE DIFFERENT

- Across **workout types and genders**, individuals between 90-110 kg tend to reach the highest max BPM across workout types
- The **overall shape** of the relationship is similar for men and women, although men show slightly more spread at higher weights.
- Strength and Yoga** reach the highest values of BPM for Females, and **Strength and HIIT** for Males

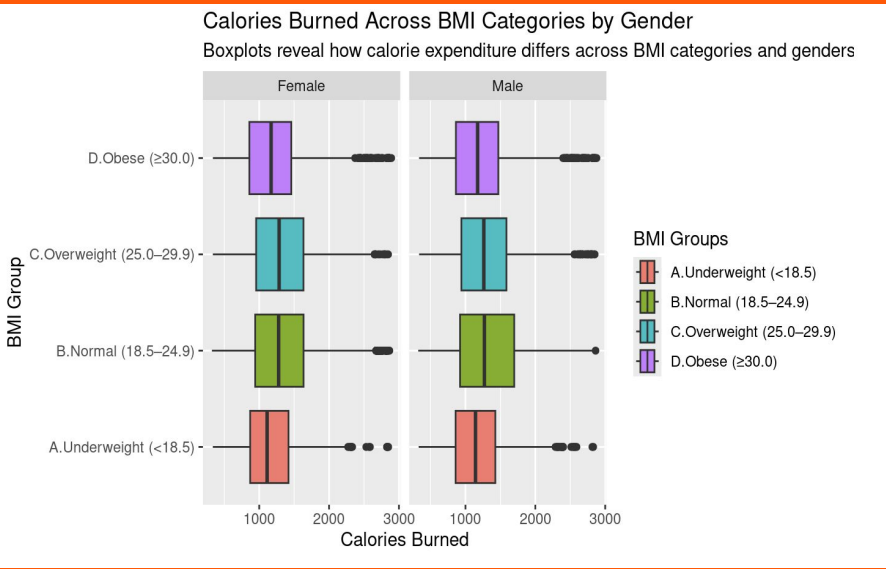


What We See:

- HIIT leads:** Median ~1,700 calories with wide range (500-3,000)
- Cardio second:** Consistent ~1,200 calories, narrow distribution
- Strength varies:** ~1,400 median but high individual variation
- Yoga lowest:** ~850 calories, most consistent results

FINDING 3: BMI Does Not Predict Calories Clearly

- Normal BMI** group shows the largest IQR, indicating the greatest central variability.
- Underweight and Obese** groups produce more outliers and have lower median calorie expenditure.
- Gender differences are small**
- When **BMI is too low or too high**, individuals tend to have more outliers and a lower median.



KEY FINDINGS:

Through these plots, we can see that, when looking at workout types and body size, the workout type factor plays a greater and more indicative role in determining the intensity of the workout and the scale of the calorie burn. One thing to note is that, across all participants, HIIT produces the highest calorie burn while yoga produces the least, however, this relation flips when looking at Max BPM across genders. For males, HIIT correlates with a greater max BPM, whereas for females, yoga does. For both calorie burn and max BPM, strength and cardio fall in the middle, indicating that they require similar, moderate energy levels. While body weight does correlate to max BPM, with individuals around 90-110 kg reaching the highest max BPM, BMI does not show a clear or consistent relationship with calories burned. Instead, calories burned vary widely within each BMI group, suggesting that factors beyond body size, such as effort level, workout type, and conditioning, are more influential. These factors indicate that it's more important to prioritize exercise type selection over physical characteristics when aiming to manage workout intensity and energy expenditure.